Supplementary Information for

Order-disorder charge density wave instability in the kagome metal (Cs,Rb)V_3Sb_5

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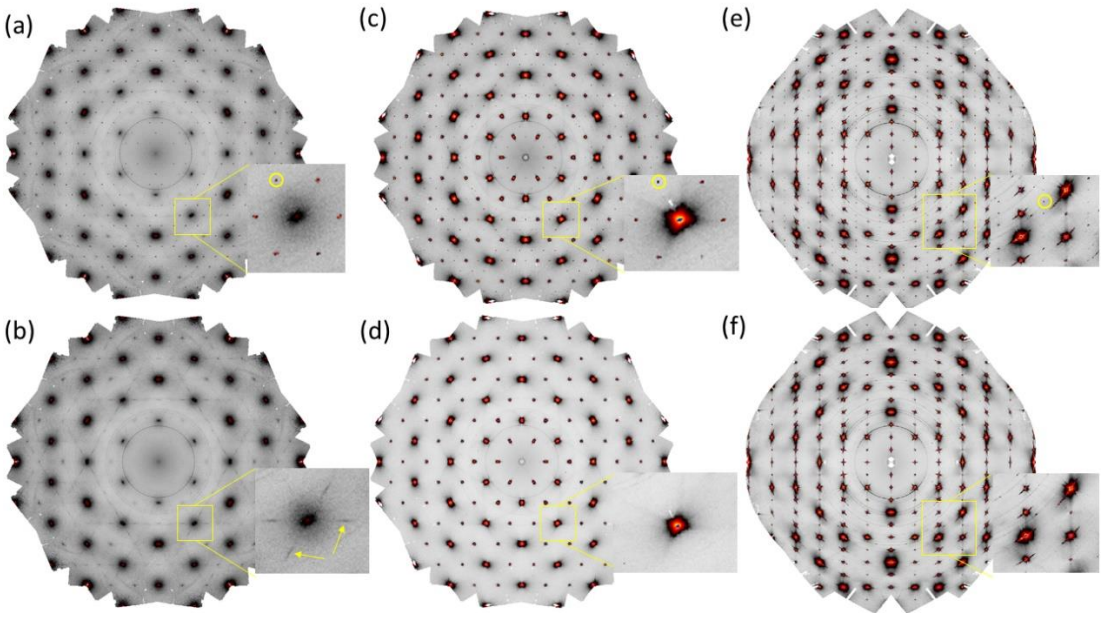
1) Diffuse scattering (DS) of RbV_3Sb_5.

We have also performed diffuse scattering on the RbV_3Sb_5 single crystals, observing the same diffuse profile as the CsV_3Sb_5 samples. The DS measurements were performed at the ID28 station at ESRF [1].

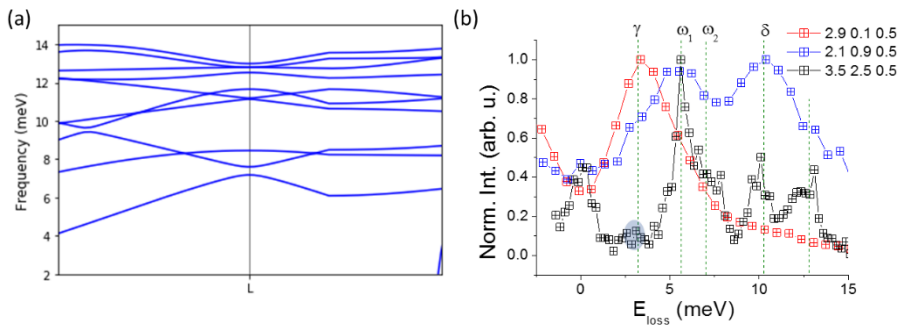


Supplementary Figure 1. Diffuse scattering at 100 K and 110 K for RbV_3Sb_5 for the (a) (*h k* 0.5) plane, showing the CDW Bragg reflections (weak spots highlighted within the yellow circle in the zoomed in area) and the precursor of the 3D CDW, marked with arrows. (b) and (c) (*h k* 0) and (*h* 0 *l*) planes, respectively (no precursor).

2) Phonon dispersion and mode assignment in CsV_3Sb_5.

Figure 2 zooms in into the low energy phonon branches around the *L* point. The calculation was carried out by means of Density Functional Perturbation Theory

(DFPT) as explained in the main text using Grimme's semiempirical approach for the van der Waals corrections [2-4].



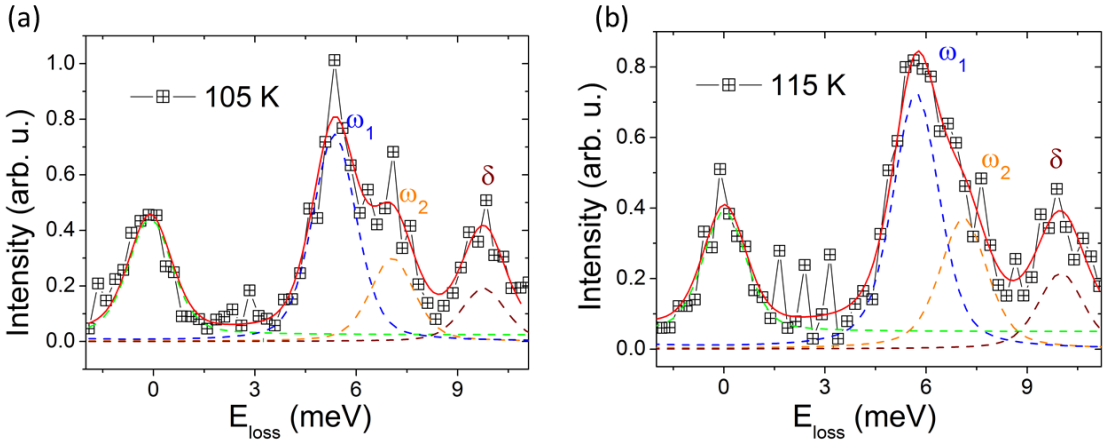
Supplementary Figure 2: (a) Zoom-in of the L point phonon dispersion of CsV_3Sb_5 calculated by Density Functional Perturbation Theory. (b) Identification of the phonon modes.

3) <u>Phonon fitting procedure</u>

Here we describe the phonon fitting procedure of figure 4(e). In the range of 12 meV there are 6 modes at the L point (see DFPT calculations in suppl. Inf. Figure 2(a)), 4 of them visible in the IXS spectra and labelled as δ, ω_1, ω_2 and δ. The two phonons ω_1 and ω_2 are very close in energy and overlap, thus a reliable fitting that reflects the experimental trend of the linewidth turns out to be a challenge.

The fitting procedure was carried out following the steps:

- First, we fit the IXS spectrum at 105 K and obtain the ω_1 and ω_2 linewidths. From this, we see that ω_1 is resolution limited.
- Looking at the raw data, both ω_1 and ω_2 increase their linewidth at 96 K and at higher temperatures as compared with 105 K scan. Trying to fit the linewidth of ω_1 and ω_2 '*from scratch*' (namely, allowing the code to reach convergence by itself) gives random values in both ω_1 and ω_2 that do not reflect the experimental trend, a T-dependent broadening.
- Therefore, we opted for fixing the linewidth of ω_2 to the value obtained at 105 K and fit ω_1 for the rest of the temperatures. The raw data shows that ω_2 broadens less than ω_1, or its broadening is merely an effect of the larger broadening experienced by ω_1. By fixing the linewidth of ω_2 during the whole fitting process, its error is zero, and this also gives an anomalously low error for ω_1.
- To circumvent this issue, now we have continued with the fitting procedure by releasing the linewidth of ω_2, thus allowing an uncertainty in ω_2, and continue with the fitting until convergence is reached.



Supplementary Figure 3: Fitting of the (a) 105 K and (b) 115 K.

We want to point out that the temperature dependence of ω_1 we show in the main text reflects a linewidth trend, as observed experimentally, and we warn the reader not to be interpreted it as the intrinsic phonon lifetime of ω_1.

References

[1] A. Girard et al., A new diffractometer for diffuse scattering studies on the ID28 beamline at the ESRF. Journal of Synchrotron Radiation, 26, 272 (2019).

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